

The Algebraic Bridge: FFGFT and $\mathbb{Z}_3\mathbb{C}$

Frobenius Automorphism, Charge Classification and
Five Structural Questions between $T^4/\mathbb{Z}_3$ and $\text{GF}(9)$

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Abstract

This document investigates five algebraic structural questions between FFGFT geometry ($T^4/\mathbb{Z}_3$, Docs. 257/321) and Matzke's GALG framework ($G(6, \mathbb{Z}_3\mathbb{C})$). Three main results **[B]**: (1) The FFGFT sector pairing $k \leftrightarrow -k$ and Matzke's $V_s = V^\dagger$ both realise the Frobenius automorphism $x \mapsto x^3$ in $\text{GF}(9)$. (2) The canonical ring homomorphism $\varphi : N \mapsto N \bmod 3$ connects Furey's number operator with the FFGFT winding quantum. (3) The injection $\iota : Q \mapsto (3Q \bmod 3, [Q])$ shows that $N \bmod 3$ fully classifies the charge $Q = N/3$.

Verification scripts: `pruef_324_z3c_trine_exakt.py`,
`pruef_325_gf9_ffgft_bruecke.py`.

.1 Motivation: Characteristic 3 and FFGFT

In characteristic 3, $(1 + N)^3 = 1$ for nilpotent N with $N^2 = 0$ holds *algebraically exactly* — because $\binom{3}{k} \equiv 0 \pmod{3}$ for $0 < k < 3$. The machine-epsilon result $|T_k^3 - \mathbf{1}| < 6.5 \times 10^{-16}$ from Doc. 324 is therefore not an approximation but a consequence of an algebraic theorem **[B]**.

.2 Five structural questions — updated status

Question 1: Number operator N and FFGFT winding quantum Δ_w

Previously open [S]; now partially closed.

Furey's number operator $N = \sum_j q_j^\dagger q_j$ has spectrum $\{0, 1, 1, 1, 2, 2, 2, 3\}$ on the eight basis states of one generation. Reduction mod 3 gives:

$$N \bmod 3 = \{0, 1, 1, 1, 2, 2, 2, 0\} \quad (1)$$

with particle classification:

$N \bmod 3 = 0$: colour-neutral (neutrino, positron)

$N \bmod 3 = 1$: anti-down quarks (3 colours)

$N \bmod 3 = 2$: up quarks (3 colours)

This is *exactly* the $\mathbb{Z}_3$ class decomposition of the FFGFT sectors $\Delta_w \in \{0, 1, 2\}$ on $T^4/\mathbb{Z}_3$.

Ring homomorphism [B]:

$$\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z}, \quad \varphi(N) = N \bmod 3 = \Delta_w \quad (2)$$

is the canonical quotient; $\varphi(N + M) = \varphi(N) + \varphi(M) \bmod 3$ is a proved algebraic property.

Status: [B] (homomorphism), [K] (physical assignment)

$\varphi(N) = N \bmod 3$ is a ring homomorphism **[B]**. The physical identification $\varphi(N) = \Delta_w$ (Furey's number operator $\leftrightarrow$ FFGFT winding quantum) classifies the same threefold particle spectrum decomposition; the deeper algebraic reason for this agreement is **[K]**.

Question 2: Charge formula $Q = N/3$ and $\mathbb{Z}_3$ classification

Previously open [S]; now partially closed.

$Q = N/3 \in \mathbb{Q}$ cannot be embedded into $\text{GF}(3)$ (3 is a zero divisor in $\mathbb{Z}/9\mathbb{Z}$, no inverse). However an injection exists:

$$\iota : \{0, \frac{1}{3}, \frac{2}{3}, 1\} \hookrightarrow \mathbb{Z}/3\mathbb{Z} \times \{0, 1\}, \quad \iota(Q) = (3Q \bmod 3, [Q]) \quad (3)$$

with values:

Q	$\iota(Q)$	Particle
0	(0, 0)	Neutrino
1/3	(1, 0)	Anti-Down
2/3	(2, 0)	Up
1	(0, 1)	Positron

The $\mathbb{Z}/3\mathbb{Z}$ component of $\iota(Q)$ is exactly $N \bmod 3$.

Status: [B] (classification), [K] (Q as $\mathbb{Q}$ number)

$N \bmod 3$ fully classifies $Q = N/3$ **[B]**. Q itself (as rational number $1/3, 2/3$) requires $\mathbb{Q}$ and is not a GF(3) object **[K]**.

Question 3: Sector pairing $k \leftrightarrow -k$ and Frobenius automorphism

Main result; was [S]; now [B].

The Frobenius automorphism in GF(9):

$$\phi : x \mapsto x^3, \quad \phi(a + bi) = a - bi \quad [\text{field conjugation}] \quad (4)$$

operates simultaneously on three levels:

- **GF(9) field:** $a + bi \mapsto a - bi$ (conjugation)
- **FFGFT sector:** $k \mapsto -k \bmod 3$ ($\mathbb{Z}_3$ inversion)
- **Matzke vacuum:** $V \mapsto V_s = V^\dagger$ (matrix adjunction)

Numerically verified: $x^3 = \bar{x}$ for all 8 non-zero GF(9) elements **[B]**.

Status [B]

All three operations are algebraically isomorphic realisations of the Frobenius automorphism $x \mapsto x^3$.

Question 4: Casimir singularity and ξ in GF(3)

$\xi = (4/3)/10^4$: Casimir denominator $3 \equiv 0$ in GF(3) is singular. $\xi \bmod 3 = 1$ — ξ lies in residue class 1. The singularity marks the transition $\text{GF}(3) \rightarrow \mathbb{R}$ **[K]**.

Question 5: $\sqrt{2}$ in GF(9) and n_{thresh}

$i^2 = -1 \equiv 2 \pmod{3}$, so $\sqrt{2} = i \in \text{GF}(9)$ **[B]**. $n_{\text{thresh}} = 2\pi/(\sqrt{2} \ln 2) \approx 6.41$ (R93) thus has an algebraically exact GF(9) kernel.

.3 Status overview

Table 1: Algebraic bridge questions FFGFT $\leftrightarrow$ GF(9) — updated status

#	Question	Result	Status
1a	Ring homo $\varphi(N) = N \bmod 3$	canonical, proved	[B]
1b	$\varphi(N) = \Delta_w$ physical	same threefold split	[K]
2a	$N \bmod 3$ classifies Q	injection ι proved	[B]
2b	$Q = N/3$ as GF(3) object	impossible ($3 \equiv 0$)	[K]
3	$k \leftrightarrow -k$ vs. $V_s = V^\dagger$	Frobenius $x \mapsto x^3$	[B]
4	$C_2 = 4/3$ and ξ in GF(3)	singularity = $\text{GF}(3) \rightarrow \mathbb{R}$	[K]
5	$\sqrt{2}$ in GF(9), n_{thresh}	$i^2 = 2$, kernel exact	[B]

Verification scripts

- `pruef_324_z3c_trine_exakt.py`: Algebraic proof $T_k^3 = \mathbf{1}$ in char. 3 **[B]**
- `pruef_325_gf9_ffgft_bruecke.py`: All five questions; Frobenius **[B]**; ring homo **[B]**; injection ι **[B]**